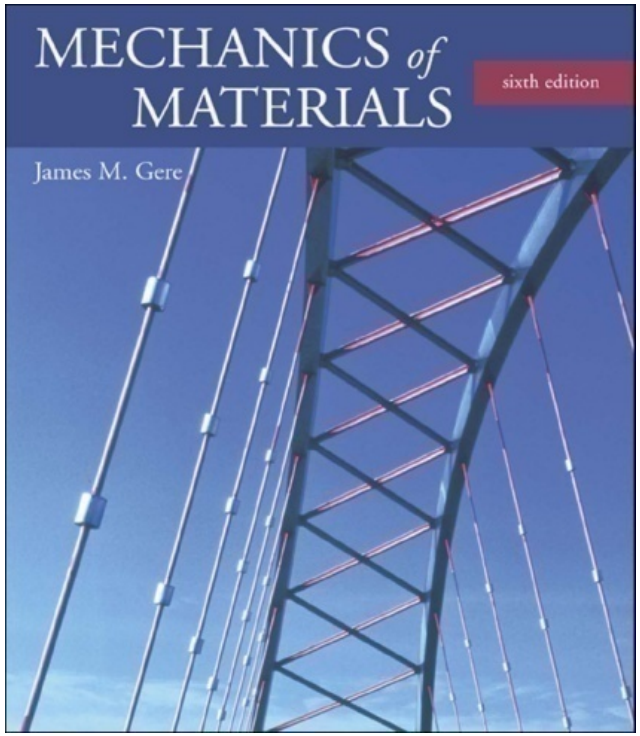
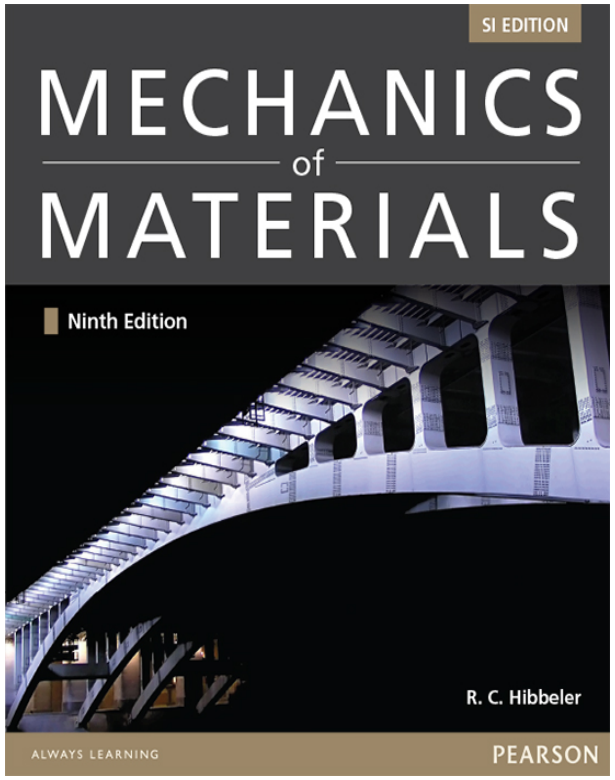
# Text books




Authors: James M. Gere
Title: Mechanics of Materials
Edition: 6th SI Edition




Authors: RC Hibbeler
Title: Mechanics of Materials
Edition: 9th SI Edition

# Basic assumption of material mechanics

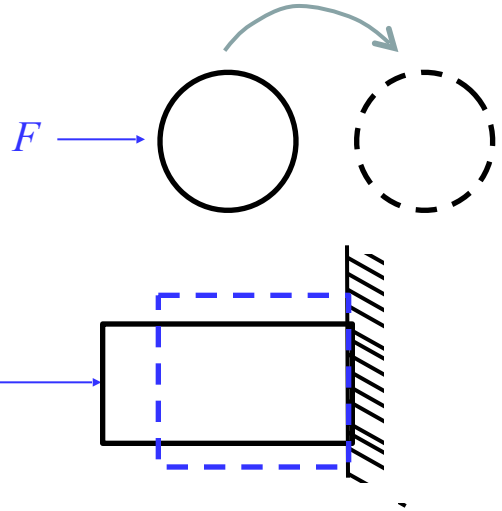
- **Continuity Assumption** — Assumes that the material is fully **filled** without voids throughout its volume, allowing mechanical quantities to be treated as continuous functions of coordinates。
- **Homogeneity Assumption** — Assumes that the mechanical properties are **identical** at every point within the material, making it macroscopically uniform。
- **Isotropy Assumption** — States that the material's mechanical properties are the same in all directions, such as elastic modulus and Poisson's ratio。
- **Small Deformation Assumption** — Assumes that deformations under load are **negligible** compared to the original dimensions, enabling analysis based on the initial geometry and linearization of equations。
- **Linear Elasticity Assumption** — Some textbooks list this as a fifth fundamental assumption, stating that within the elastic range, stress is **proportional** to strain, and the material fully returns to its original shape after unloading.

As the name implies, Mechanics of Materials is about the mechanical behavior of materials due to external forces. It studies the ***Theory of Deformable Bodies***.

**Difference between *Fundamentals of Mechanics* and *Mechanics of Materials***

- *Fundamentals of Mechanics*: deals with ideal, rigid bodies that are non-deformable.

- *Mechanics of Materials*: deals with real, deformable materials/structures.





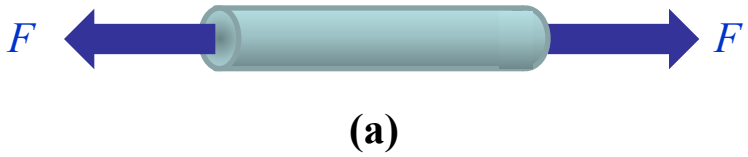



http://www.strand7.com/html/archive/structural/building.htm

# Chapter 1 – Stress

## 1.1 Introduction

Consider two components (a and b) made of the same material, an equal force ($F$) is applied to each of the component. If the force $F$ is steadily increased, which component will break first?



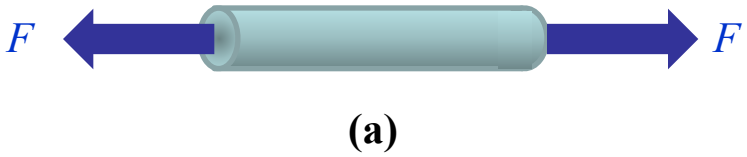

**(a)**




**(b)**

--- the thinner one (a).
Because the force is distributed over a smaller cross-sectional area.

Consider two equal components (a and b) made of the same material, unequal forces ($F$ and $2F$) are applied to them. If force $F$ and $2F$ are steadily increased, which component will break first?




**(a)**




**(b)**

--- the one subjected to larger force, (b).
Because the intensity of the force is bigger.

**Which one breaks first ?**




**(a)**

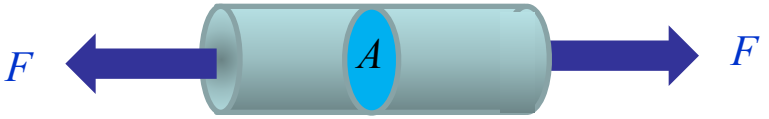



**(b)**

--- I do not know.

(a): thinner but the applied force is smaller,

(b): thicker but the applied force is also larger.




cross-sectional area: The quantity of matters.

strength : the intensity of the force

How do you stop something from breaking?

1. Make it out of bigger components — change its cross-sectional area
2. Make it out of a stronger material — change its strength
3. Avoid strong external forces.

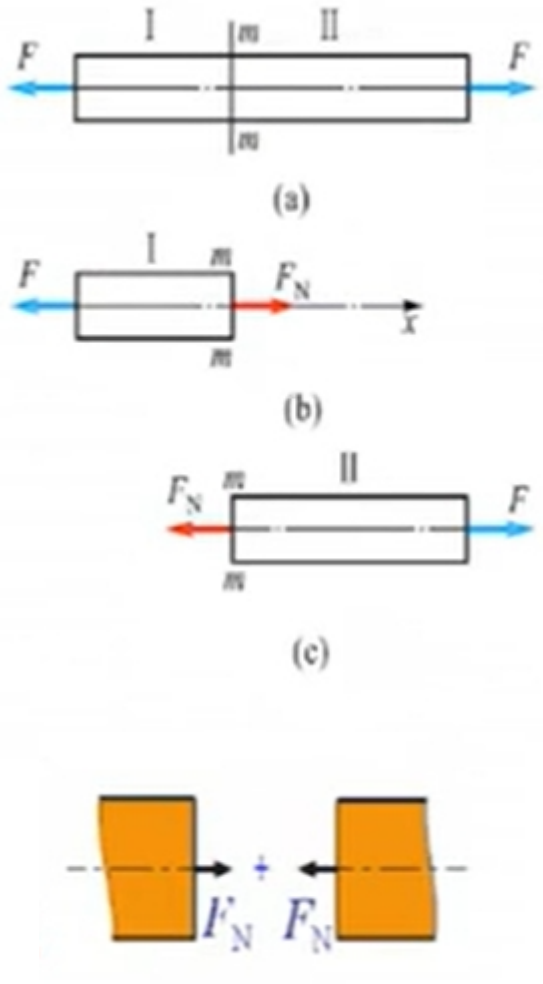
**Solution procedure for average normal stress for a given structure:**

**1)** Divide the structure into some regions and determine the ***internal*** normal forces $F$ for each region from the equilibrium condition of the region using free body diagram

**2)** Determine the cross-sectional area: $A$

**3)** Calculate average normal stress as $F/A$

## 1.3 Internal forces in a body




**External forces:** Forces applied to a body. They are resulted from the interaction between body and its environment. In this problem, the external forces is $F$.

The external forces are positive if they point to a positive coordinate axis. They are negative if they point to a negative coordinate axis.

For external moments (couples), they can be assumed as positive if they are anticlockwise, and negative if they are clockwise.

**Internal forces:** Forces act from within a body (forces the body carries). In this problem, the internal forces are $F_N$.

The internal forces are assumed to be positive if they are tensile. They are negative if they are compressive.

## 1.2 Normal stress

*F* is internal force of the cross-section of the body

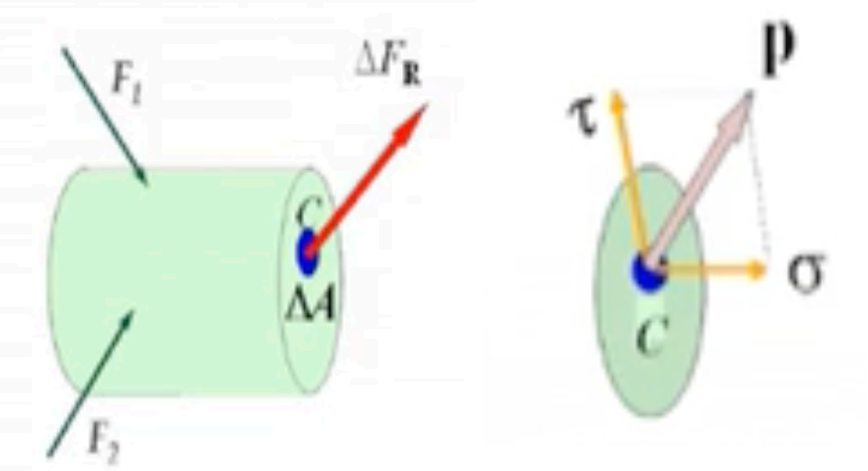
**Sign Conventions:**

- *F* is assumed to be positive if it is tensile.
- *F* is assumed to be negative if it is compressive.

The average intensity of internal force - **average stress**:

$$p_a = \frac{\Delta F_R}{\Delta A}$$

Take limit: $p = \lim_{\Delta A \to 0} p_a = \lim_{\Delta A \to 0} \frac{\Delta F_R}{\Delta A}$




*P* is the intensity of the internal force acting on the unit area
*P* is composed of σ (along axis) and τ in the cross section
σ (sigma) is a Greek letter

- **Stress**: The intensity of the internal force acting on a specific area of per unit.

Decompose $p = \sigma + \tau$

- **Normal stress**: The intensity of the force acting **normal** to the cross-sectional area

Normal Stress

Shear Stress

**Units of the stress**

$$\textit{Units of}\ \text{normal stress} = \frac{\text{Unit of force}}{\text{Unit of area}} = \frac{\text{N}}{\text{m}^2} = \text{Pa}$$

Typical values of normal stress are commonly measured in:

MPa ($10^6$ Pa, Mega-pascal)

or

GPa ($10^9$ Pa, Giga-pascal)

**Sign Conventions:**

- $\sigma$ is positive if it is tensile.
- $\sigma$ is negative if it is compressive.

For your information, atmospheric pressure (at sea-level) is: 101325 Pa or 0.101325 MPa

# 1.4 Shear stress

**Shear Stress**: The intensity of the force acting **parallel** to the cross-sectional area



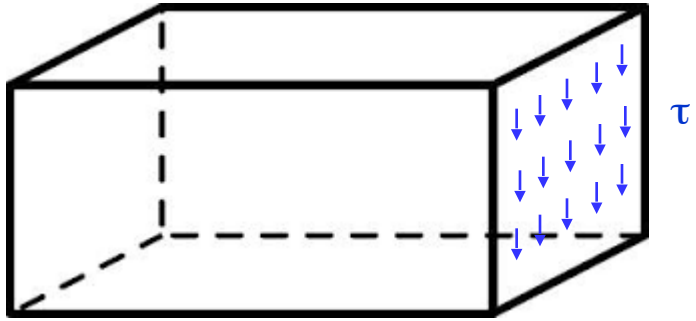

**(Average) shear stress:**

$\tau$ (tao) is a Greek letter

$F$ is the ***internal*** shear force

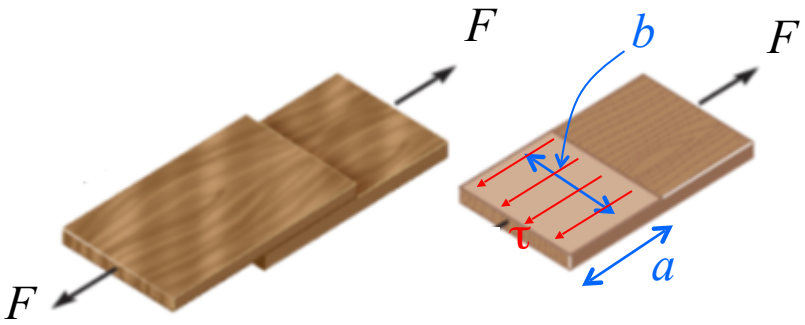
$$\tau = F / A$$

Unit of shear stress




MPa and GPa are commonly used.

## Another example of shear stress




Two plates are glued together.

The overlapping area is $a \times b$.

Shear stress : $\tau = F/A = F/(ab)$

## READING QUIZZES

1. What is the average normal stress in the bar for loading cases A and B if $P = 10$ kN and $A = 500$ mm²?

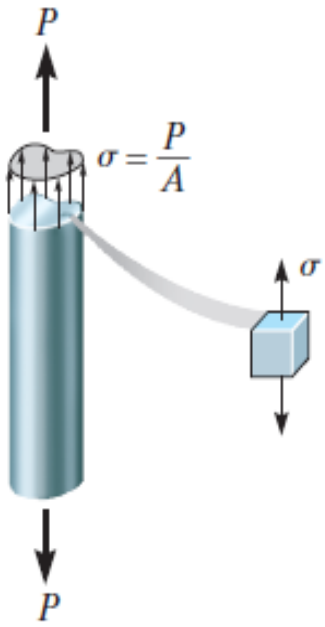
a) A: 0.02 kPa; B: –0.02 kPa

b) A: 20 Pa; B: –20 Pa

c) A: 20 MPa; B: 20 MPa

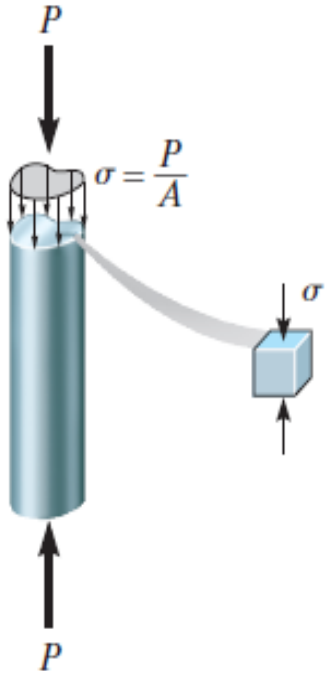
d) A, B: 200 N/mm²

e) A: 20 MPa; B: –20 MPa




Tension

A




Compression

B

Answer: e

2. What is the average shear stress in the internal vertical surface AB (or CD), if $F$ = 20 kN, and $A_{AB} = A_{CD}$ = 1,000 mm²?

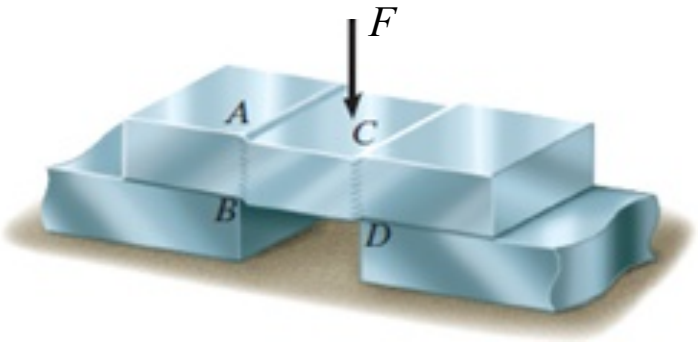
a) 20 N/mm²

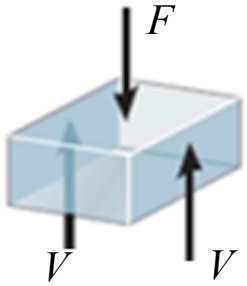
b) 10 N/mm²

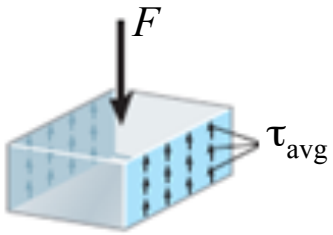
c) 10 kPa

d) 200 kN/m²

e) 20 MPa



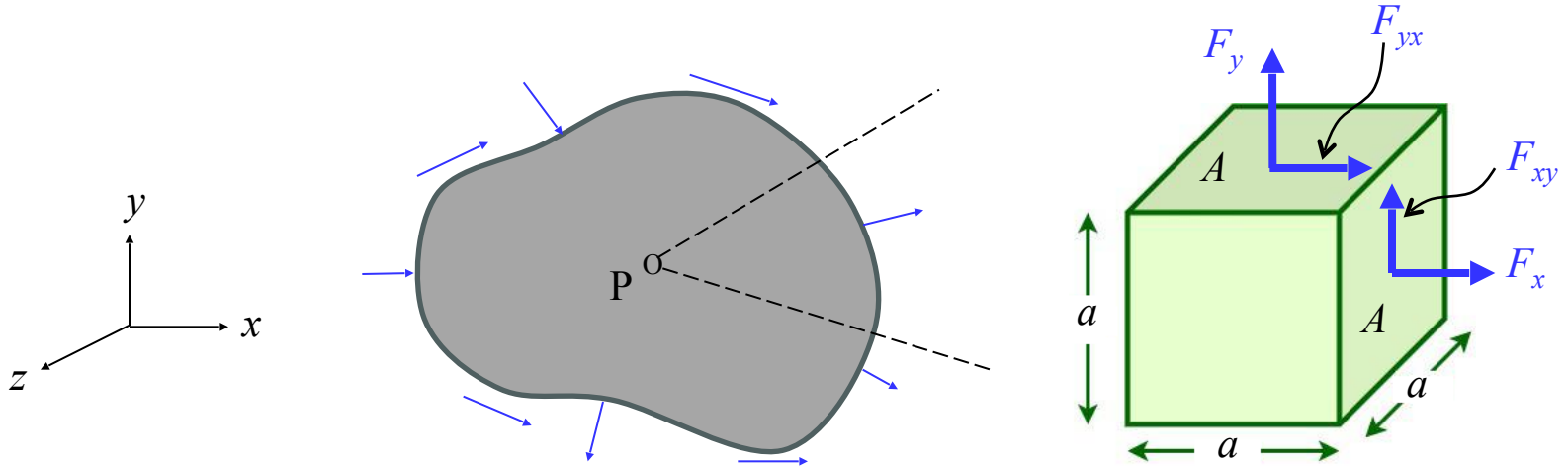



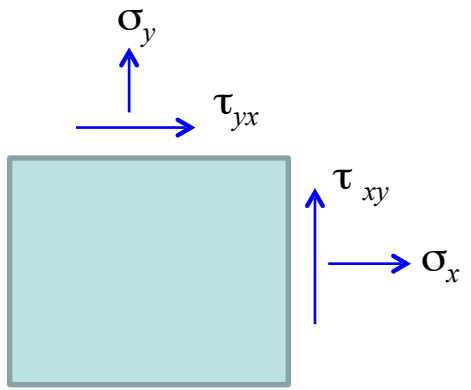

Answer: b

# 1.5 Stresses under general loads

Consider a body with multiple loads applied to it. To define the stress at a point P, we draw a very small cubic at P. We then magnify the cubic and look at the forces that the adjoining material applies to the cubic along each of its faces.

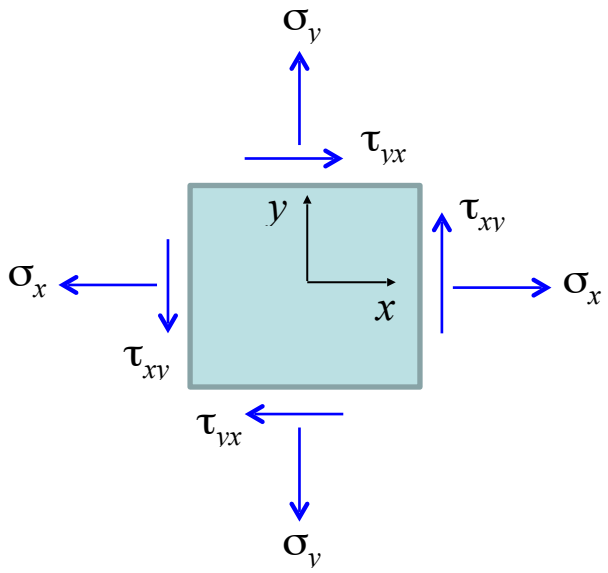



We only consider the forces in *x* and *y* directions as shown. The stresses on each face are obtained by dividing the forces by the area of the face. For example, on the face pointing in the positive *x* direction,




And on the face pointing in the positive *y* direction,

*A* is the area of each face of the cubic.




**The stresses are positive if either:**
1) The direction of the face and the direction of the force are both positive or
2) The direction of the face and the direction of the force are both negative

**The stresses are negative if either:**
1) The direction of the face is positive and the direction of the force is negative or
2) Vice Versa.

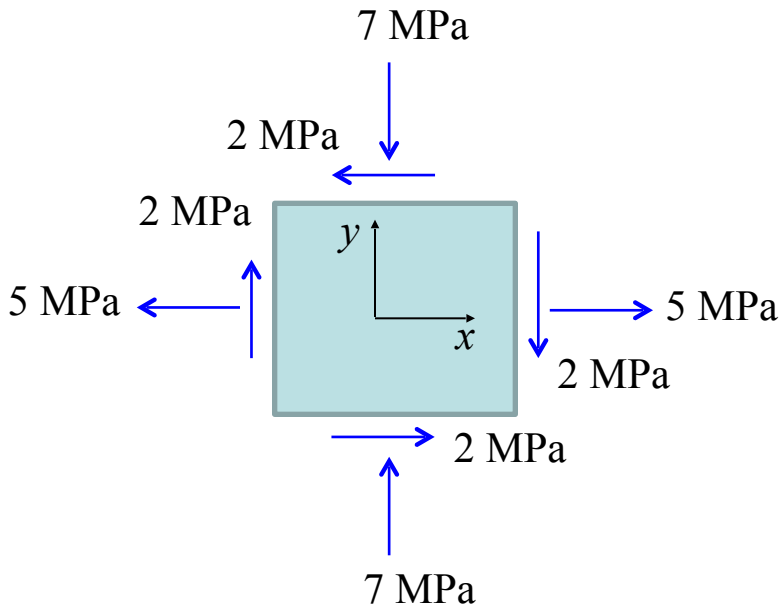
• We use the subscripts to distinguish between the normal stresses acting on the *x* and *y* faces ($\sigma_x$ and $\sigma_y$). The normal stresses are positive if they are tensile. The normal stresses are negative if they are compressive.

• For the shear stress on the positive *x* face, the stress acts in the *y* direction. We therefore use double index notation to label this shear stress $\tau_{xy}$. The first subscript refers to the direction of the face and the second subscript refers to the direction in which the stress acts.

• Since the square is very small, the stresses acting on the positive and negative *x* faces are equal, and likewise, for the positive and negative *y* faces.

• The stresses in the figure above are all shown in the directions that we define to be positive.

**Example:**



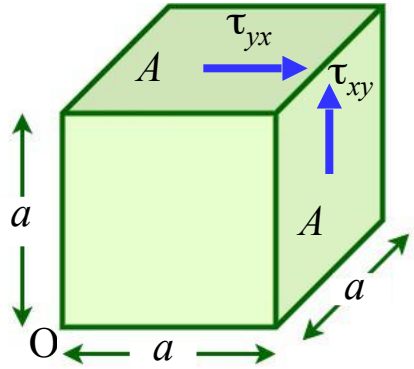

$\sigma_x = 5$ MPa

$\sigma_y = -7$ MPa

$\tau_{yx} = -2$ MPa

$\tau_{xy} = -2$ MPa

## 1.6. Symmetry of shear stress components




*Now, expressing the stress components as forces, and summing moments about the low-left corner of the cubic (point O).

Total torque = 0.

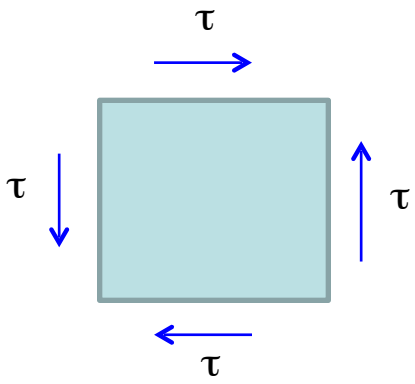
$\tau_{xy} \times A \times a - \tau_{yx} \times A \times a = 0$

$A$ is the area of the surfaces of the cubic.

Hence, $\tau_{xy} = \tau_{yx}$

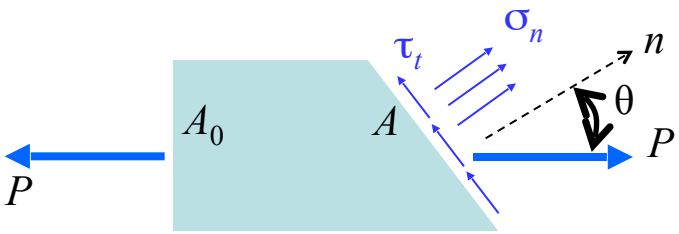
This saves us from having to remember the difference between $\tau_{xy}$ and $\tau_{yx}$.

In two-dimensional situation, we can simply use the symbol $\tau$ to denote the shear stress.



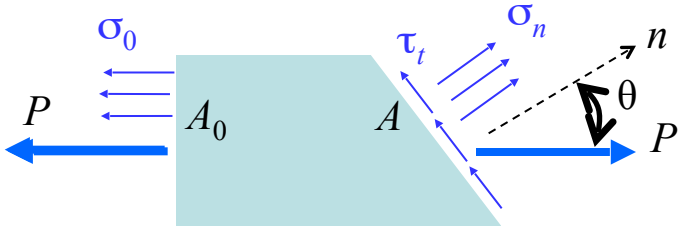
## 1.7 Stress on an oblique plane

In some situations we want to find the stress on a plane, which is neither normal to nor parallel to the applied force.




Normal stress on $A_0$ ($\sigma_0$) :

$$\sigma_0 = P / A_0$$




Normal stress on $A$ ($\sigma_n$) :

$$\sigma_n = P_n / A$$
$$= P\cos\theta / A$$

Shear stress on $A$ ($\tau_t$) :

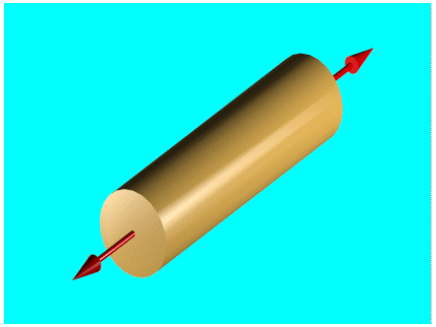
$$\tau_t = P\sin\theta / A$$

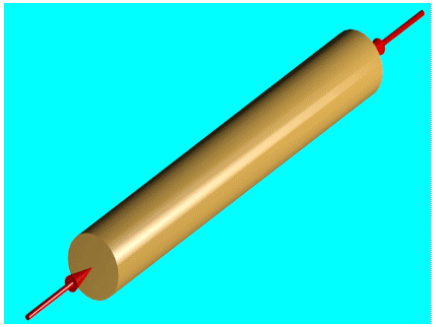
# Chapter 2 – Strain

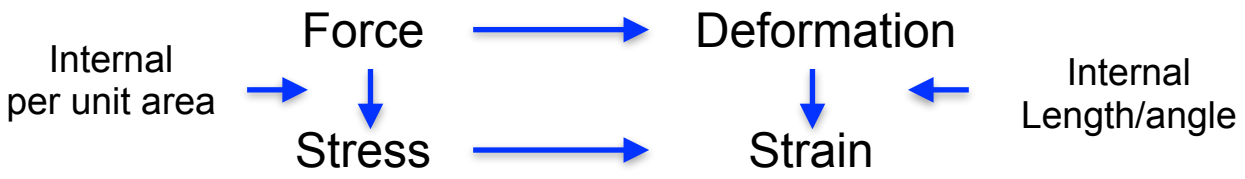
## 2.1 Introduction

In *Mechanics of Materials*, we remove the assumption in the subject *Fundamentals of Mechanics* that all structures are rigid bodies.

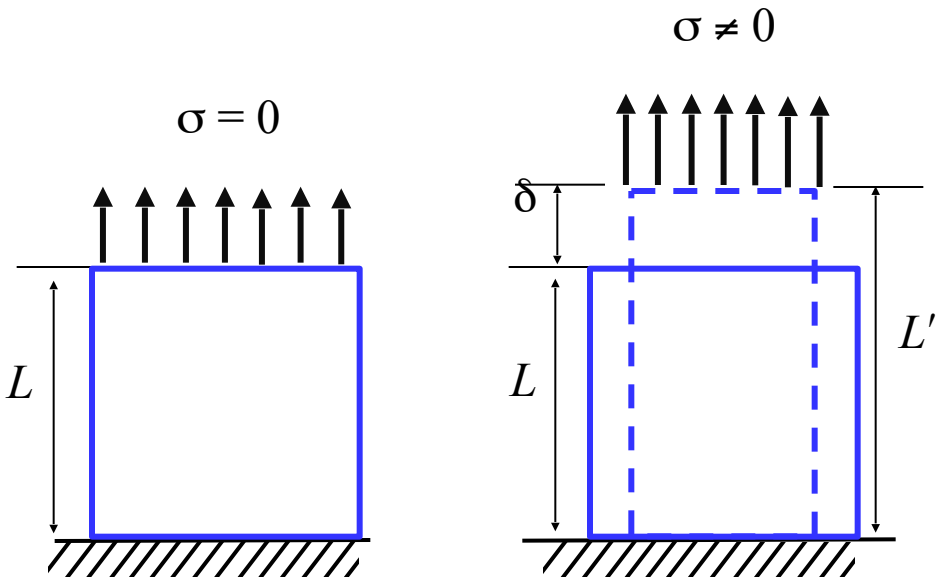
Instead, under external loads, the structure will deform **(size change or shape change)**







## 2.2 Normal strain




Under the action of a positive normal stress (σ), a square (solid) lengthens by an amount δ and becomes a rectangular (dashed).

Normal stress  Deflection, elongation

σ ⇨ δ

Average normal stain of the square:

$$\varepsilon = \delta / L$$

$$\delta = L' - L$$

$$\varepsilon = (L' - L)/L$$

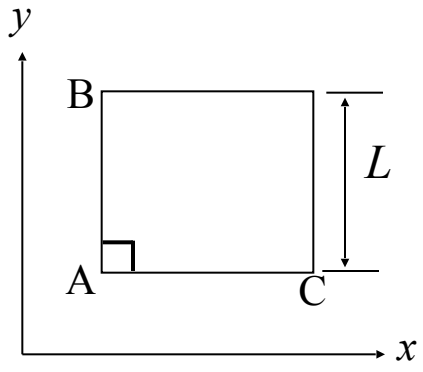
- If the square becomes longer, the normal strain is positive.
- If the square becomes shorter, the normal strain is negative.

Pronounce δ : delta

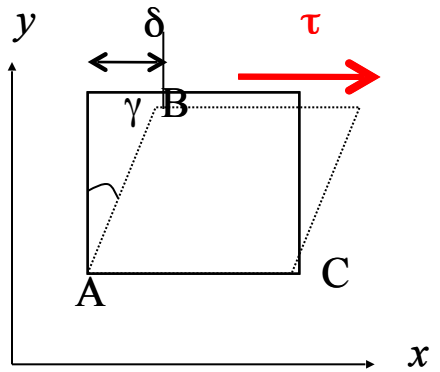
Pronounce ε : Epsilon

The units of normal stain are:

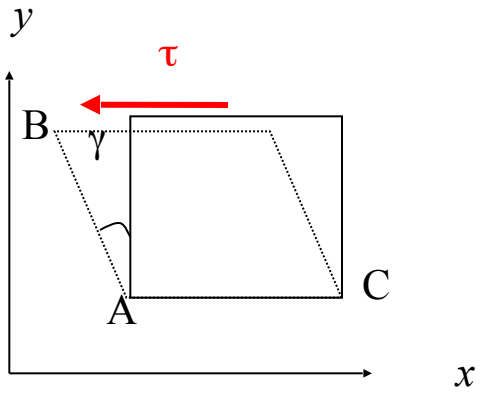
## 2.3 Shear strain




The square becomes a diamond when it is loaded by a shear stress τ.




Positive shear stress, positive shear strain, $\gamma > 0$, angle BAC reduces.



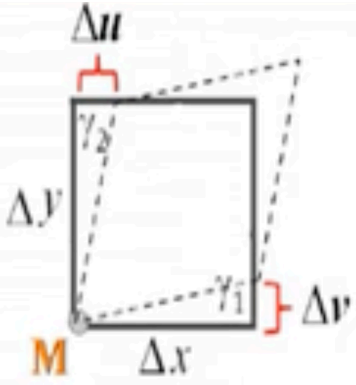

Negative shear stress, negative shear strain, $\gamma < 0$, angle BAC increases.

Deflection of the top surface of the square (δ) is in direction *x*.

Pronounce γ: gama

Shear strain ( γ ): Change in angle with respect to *x* and *y* axes (measured in radians)

Units of shear strain = Units of angle = Radians (or dimensionless)

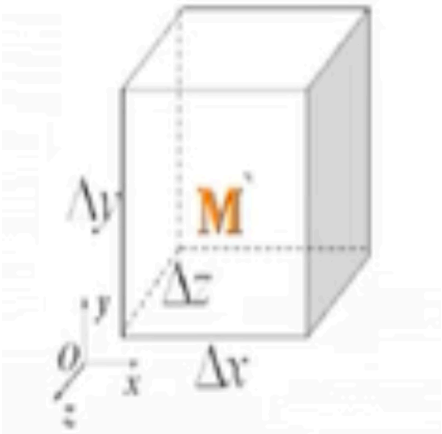



Shear strain $\gamma_{xy} = \gamma_1 + \gamma_2 = \lim_{\Delta x, \Delta y \to 0} \left( \frac{\Delta v}{\Delta x} + \frac{\Delta u}{\Delta y} \right)$

$(\gamma_{xy} = \gamma_{yx})$

Same as:

$\gamma_{yz} = \gamma_{zy};\ \gamma_{zx} = \gamma_{xz}$




Strain status at a point:

$(\varepsilon_x, \gamma_{xy}, \gamma_{xz};\ \gamma_{yx}, \varepsilon_y, \gamma_{yz};\ \gamma_{zx}, \gamma_{zy}, \varepsilon_z)$